

CBSE Class 12 Physics — Dual Nature of Radiation and Matter

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. A metallic plate exposed to white light emits electrons. For which of the following colours of light will the stopping potential be strictly maximum? (a) Blue (b) Yellow (c) Red (d) Violet.

[PYQ 2023] Very High

Q2. Which one of the following elements will require the highest energy to take out an electron from its surface? (a) Ge (b) C (c) Si (d) Pb. [PYQ 2023] Very High

Q3. A photon of wavelength **663 nm** is incident on a metal surface. The work function of the metal is **1.50 eV**. The maximum kinetic energy of the emitted photoelectrons is: (a) $3.0 \times 10^{-20} \text{ J}$ (b) $6.0 \times 10^{-20} \text{ J}$ (c) $4.5 \times 10^{-20} \text{ J}$ (d) $9.0 \times 10^{-20} \text{ J}$. [PYQ 2023] High

Q4. The work function for a given metal surface is **4.14 eV**. The threshold wavelength for this metal surface is: (a) **4125 \AA** (b) **2062.5 \AA** (c) **3000 \AA** (d) **6000 \AA**. [PYQ 2022] Very High

Q5. A photocell connected in an electrical circuit is placed at a distance **d** from a source of light. As a result, a current **I** flows in the circuit. What will be the current in the circuit when the distance is reduced exactly to **$d/2$** ? (a) **I** (b) **$2I$** (c) **$4I$** (d) **$I/2$** . [PYQ 2020] Very High

Q6. If photons of frequency **ν** are incident on the surfaces of metals A and B of threshold frequencies **$\nu/2$** and **$\nu/3$** respectively, the ratio of the maximum kinetic energy of electrons emitted from A to that from B is: (a) **2 : 3** (b) **3 : 4** (c) **1 : 3** (d) **$\sqrt{3} : \sqrt{2}$** . [PYQ 2020] High

Q7. The kinetic energy of a proton and that of an **α** -particle are **4 eV** and **1 eV**, respectively. The ratio of the de-Broglie wavelengths associated with them will be: (a) **2 : 1** (b) **1 : 1** (c) **1 : 2** (d) **4 : 1**. [PYQ 2020] Very High

Q8. What is the exact wavelength of a photon having an energy of **$3.3 \times 10^{-19} \text{ J}$** ? [PYQ 2020] Medium

Q9. In the wave picture of light, the intensity of light is determined by the square of the amplitude of the wave. What specifically determines the intensity of light in the photon (quantum) picture of light? [PYQ 2022] Very High

Q10. Define the term "threshold frequency" in the context of photoelectric emission. [PYQ 2019] Very High

Q11. Define the term "stopping potential" in the study of the photoelectric effect. [PYQ 2019] Very High

Q12. Assertion (A): For radiation of a frequency greater than the threshold frequency, photoelectric current is directly proportional to the intensity of the radiation. **Reason (R):** Greater the number of energy quanta available, greater is the number of electrons absorbing the energy quanta and greater is the number of electrons coming out of the metal. [PYQ 2023] Very High

Q13. Assertion (A): The photoelectrons produced by a monochromatic light beam incident on a metal surface have a spread in their kinetic energies. **Reason (R):** The energy of electrons emitted from deep inside the metal surface is lost in collision with the other atoms in the metal. [PYQ 2022] High

Q14. Assertion (A): The work function varies from one photosensitive metal to another. **Reason (R):** Work function fundamentally depends on the nature of the material and the properties of its surface. [Practice] High

Q15. Name the physical phenomenon which directly demonstrates the quantum (particle) nature of electromagnetic radiation. [PYQ 2017] Very High

Q16. What is the defined rest mass of a photon? [Practice] Medium

Q17. Does the stopping potential in a photoelectric setup depend on the intensity of the incident light? [PYQ 2014] Very High

Q18. An electron and a proton are moving and have the exact same kinetic energy. Which one of the two has a greater de Broglie wavelength? [Practice] High

Q19. Write the mathematical relation for the de Broglie wavelength λ of an electron accelerated from rest through a potential difference V . [Practice] Very High

Q20. If the operating frequency of incident light in a photoelectric experiment is doubled, keeping the intensity entirely constant, what happens to the saturation photoelectric current? [PYQ 2020] High

2-Mark Questions

Q21. What is the exact effect on the (i) threshold frequency, and (ii) stopping potential, when the frequency of the incident beam of light is increased? Justify your answer. [PYQ 2023] High

Q22. Why is it strictly the frequency, and not the intensity of the light source, that determines whether the emission of photoelectrons will occur or not? Explain. [PYQ 2022] Very High

Q23. If the frequency of light incident on the cathode of a photocell is systematically increased, how will the following be affected? (A) Maximum kinetic energy of photoelectrons. (B) Photocurrent. Justify. [PYQ 2020] Very High

Q24. An α -particle and a proton are accelerated from rest through the same potential difference V . Find the exact ratio of their respective de Broglie wavelengths. [PYQ 2017] Very High

Q25. Light of the exact same wavelength is incident on three photosensitive surfaces A, B, and C. The following observations are recorded: (I) From surface A, no photoelectrons are

emitted. (II) From surface B, photoelectrons are just barely emitted. (III) From surface C, photoelectrons with some kinetic energy are emitted. Compare the threshold frequencies of the three surfaces and justify your answer. [PYQ 2022] High

Q26. Monochromatic light of frequency $6.0 \times 10^{14} \text{ Hz}$ is produced by a laser. The power emitted is $2.0 \times 10^{-3} \text{ W}$. Calculate the number of photons emitted on an average per second by the source. [PYQ 2018] High

Q27. Calculate the de Broglie wavelength associated with a cricket ball of mass 150 g moving at a speed of 30 m/s . [Practice] Medium

Q28. The figure shows the stopping potential (V_0) versus $1/\lambda$ graph for two metals A and B. Write the equation relating these variables and explain how the value of Planck's constant can be experimentally determined from the slope of this graph. [PYQ 2020] Very High

Q29. An electron microscope fundamentally uses electrons accelerated by a voltage of 50 kV . Determine the de-Broglie wavelength associated with these electrons. [PYQ 2014] High

Q30. Plot a schematic graph showing the variation of de-Broglie wavelength λ versus $1/\sqrt{V}$, where V is the accelerating potential for a charged particle. How does the slope of this graph give information regarding the specific charge (q/m) of the particle? [PYQ 2019] High

Q31. Two monochromatic radiations of frequencies ν_1 and ν_2 ($\nu_1 > \nu_2$) and having the exact same intensity are in turn incident on a photosensitive surface. Explain, giving mathematical reasons, in which case the maximum kinetic energy of the emitted photoelectrons will be more. [PYQ 2014] Very High

Q32. A proton and an α -particle have the exact same de-Broglie wavelength. Determine the ratio of their respective speeds. [PYQ 2015] Medium

3-Mark Questions

Q33. Photoelectric emission occurs when a given surface is irradiated with radiation of frequency ν_1 and ν_2 . The maximum kinetic energy of the electrons emitted in the two cases are exactly K and $2K$ respectively. Obtain the algebraic expression for the threshold frequency (ν_0) of the material. [PYQ 2022] Very High

Q34. The wavelength λ of a photon and the de-Broglie wavelength of an electron of mass m have the exact same value. Show mathematically that the energy of the photon is strictly $\frac{2\lambda mc}{h}$ times the kinetic energy of the electron. [PYQ 2023] High

Q35. Explain, how does (A) the photoelectric current and (B) the maximum kinetic energy of the photoelectrons emitted in a photocell vary if the frequency of the incident radiation is strictly doubled, while keeping the incident intensity exactly the same? Show the graphical variation in the above two cases. [PYQ 2022] Very High

Q36. The work function of a metal is 2.31 eV . Photoelectric emission occurs when light of frequency $6.4 \times 10^{14} \text{ Hz}$ is incident on the metal surface. Calculate: (A) the energy of the incident radiation, (B) the maximum kinetic energy of the emitted electron, and (C) the stopping potential of the surface. [PYQ 2022] High

Q37. The maximum kinetic energy of the photoelectrons emitted is perfectly doubled when the wavelength of light incident on a photosensitive surface changes from λ_1 to λ_2 . Deduce the formal expressions for the threshold wavelength (λ_0) and the work function (Φ) for the metal surface in terms of λ_1 and λ_2 . [PYQ 2020] High

Q38. Why is the classical wave theory of electromagnetic radiation practically unable to explain the photoelectric effect? Explain with the help of Einstein's photoelectric equation any two observed features in the photoelectric effect which cannot be explained by classical wave theory. [PYQ 2019] Very High

Q39. If light of wavelength **412.5 nm** is incident on each of the metals given below, which of them will actively show photoelectric emission and why? Na ($W = 1.92 \text{ eV}$), K ($W = 2.15 \text{ eV}$), Ca ($W = 3.20 \text{ eV}$), Mo ($W = 4.17 \text{ eV}$). [PYQ 2018] Very High

Q40. State Einstein's photoelectric equation explaining all the symbols used. A graph of the square of the maximum speed of the electrons (v_{max}^2) versus incident frequency ν is obtained. Using Einstein's equation, obtain expressions for (A) Planck's constant, and (B) the work function of the material in terms of the slope (l), intercept (n), and mass of the electron (m). [PYQ 2018] High

Q41कायदा. A beam of monochromatic radiation is steadily incident on a photosensitive surface. Answer the following questions giving strict scientific reasons: (A) Do all the emitted photoelectrons have the exact same kinetic energy? (B) Does the maximum kinetic energy of the emitted electrons depend on the intensity of radiation? (C) On what specific factors does the number of emitted photoelectrons per second depend? [PYQ 2015] Very High

Q42. Find the frequency of light which ejects electrons from a metal surface, fully stopped by a retarding potential of **3.3 V**. If photoelectric emission barely begins in this metal at a frequency of $8 \times 10^{14} \text{ Hz}$, also calculate the exact work function (in eV) for this metal. [PYQ 2018] High

5-Mark Questions (Long Answer/Case-Study)

Q43. Case Study: Characteristics of Photoelectric Effect Figure shows the variation of photoelectric current measured in a photocell circuit as a function of the potential difference between the plates when light beams A, B, C, and D of different wavelengths and intensities are incident on the photocell. (A) Comparing beams A and B (which share the same stopping potential but different saturation currents), which beam has higher intensity and why? (B) Which of the four light beams has the highest frequency and why? (C) Which light beam ejects photoelectrons with the maximum linear momentum and why? (D) Describe the exact effect on the stopping potential if the intensity of beam A is doubled without changing its frequency. [PYQ 2023] Very High

Q44. (A) Using the photon picture of light, show systematically how Einstein's photoelectric equation can be established. (B) Write three basic properties of photons which are fundamental to obtaining this equation. (C) Radiation of frequency 10^{15} Hz is incident on two photosensitive surfaces P and Q. There is absolutely no photoemission from surface P.

Photoemission occurs from surface Q, but the photoelectrons have strictly zero kinetic energy. Explain these observations physically and find the exact value of the work function for surface Q (in eV). [PYQ 2017 / 2014] Very High

Q45. (A) What is the de Broglie hypothesis? Write its mathematical formula. (B) Plot a graph showing the variation of de-Broglie wavelength λ versus $1/\sqrt{V}$, where V is the accelerating potential, for two particles A and B carrying the same charge but having masses m_1 and m_2 ($m_1 > m_2$). Which line represents the particle of smaller mass and why? (C) A proton and an alpha particle have the exact same kinetic energy. Which one has a shorter de Broglie wavelength? Prove it mathematically. [PYQ 2016 / 2019] High

Q46. (A) Define the terms threshold frequency and stopping potential in relation to the photoelectric effect. (B) Light of wavelength **2000 Å** falls on a metal surface of work function **4.2 eV**. (i) What is the maximum kinetic energy (in eV) of the fastest electrons emitted from the surface? (ii) What will be the change in the maximum kinetic energy of the emitted electrons if the intensity of light with the same wavelength is doubled? (iii) If the exact same light falls on another surface of work function **6.5 eV**, what will be the energy of the emitted electrons? [PYQ 2022 / 2019] Very High

Q47. Case Study: Particle Nature of Light The particle nature of light was established by the photoelectric effect and Compton scattering. According to this picture, light interacts with matter as discrete packets of energy called photons. (A) Explain how the macroscopic "intensity" of light is interpreted at the microscopic level in the photon picture. (B) Does the rest mass of a photon physically exist? What is the algebraic expression for the momentum of a photon of frequency ν ? (C) Monochromatic light of frequency **6.0×10^{14} Hz** is produced by a laser. The power emitted is **2.0×10^{-3} W**. Calculate the exact energy of a single photon in the light beam, and the total number of photons emitted on average per second. [PYQ 2018 / Practice] High

Q48. (A) Draw a neatly labelled graph showing the variation of photoelectric current with applied voltage for two incident radiations of equal frequency but different intensities. (B) Draw a graph showing the variation of stopping potential with the frequency of incident radiation for two photosensitive metals A and B having work functions $\Phi_A > \Phi_B$. How can the work function be extracted from this graph? (C) The threshold frequency of a metal is f . When light of frequency **$2f$** is incident on the metal plate, the maximum velocity of photoelectrons is v_1 . When the frequency of incident radiation is increased to **$5f$** , the maximum velocity of photoelectrons becomes v_2 . Find the exact numerical ratio $v_1 : v_2$. [PYQ 2018 / 2016] Very High

Q49. (A) Obtain the mathematical relation between the de-Broglie wavelength (λ) and the kinetic energy (K) of a moving material particle of mass m . (B) An electron, an alpha-particle, and a proton have the exact same kinetic energy. Which of these particles has the largest de-Broglie wavelength and why? (C) The wavelength of light from the spectral emission line of sodium is **590 nm**. Find the kinetic energy (in Joules and eV) at which an electron would have the exact same de-Broglie wavelength. [PYQ 2019 / Practice] High

Q50. (A) Explain the experimental setup of the photoelectric effect. How does the experimentally observed stopping potential vary with the intensity of incident light? (B) The work functions of the following metals are given: Na = **2.75 eV**, K = **2.3 eV**, Mo = **4.17 eV**, and Ni = **5.15 eV**. Which of these specific metals will *not* cause photoelectric emission for radiation of wavelength **3300 Å** from a laser source placed **1 m** away? (C) What happens to the emission status if the laser source is brought nearer and placed **50 cm** away? Justify. [PYQ 2017 / 2015] Very High

Answer Key

Q1: (d) Violet. (Violet light has the highest frequency among visible colors, yielding the highest maximum kinetic energy and thus the maximum stopping potential). **Q2:** (b) C. (Carbon has the highest ionization energy / work function among the given group 14 elements). **Q3:** (b) 6.0×10^{-20} J. ($E = \frac{1240}{663} \approx 1.87$ eV. $K_{max} = 1.87 - 1.50 = 0.37$ eV. In Joules: $0.37 \times 1.6 \times 10^{-19} \approx 6.0 \times 10^{-20}$ J). **Q4:** (c) **3000 Å**. ($\lambda_0 = \frac{hc}{W} = \frac{1242 \text{ eV nm}}{4.14 \text{ eV}} = 300 \text{ nm} = 3000 \text{ Å}$). **Q5:** (c) **4I**. (Intensity $I \propto 1/d^2$. If distance d is halved, intensity becomes 4 times, so current becomes **4I**). **Q6:** (b) **3 : 4**. ($K_A = h\nu - h(\nu/2) = 0.5h\nu$. $K_B = h\nu - h(\nu/3) = \frac{2}{3}h\nu$. Ratio = $0.5/(2/3) = 3 : 4$). **Q7:** (b) **1 : 1**. ($\lambda = \frac{h}{\sqrt{2mK}}$. Ratio $\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha K_\alpha}{m_p K_p}} = \sqrt{\frac{4m_p \times 1}{m_p \times 4}} = 1 : 1$). **Q8:** **600 nm**. ($E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{3.3 \times 10^{-19}} = 6 \times 10^{-7} \text{ m} = 600 \text{ nm}$). **Q9:** It is determined by the number of photons crossing a unit area per unit time. **Q10:** The minimum frequency of incident radiation below which photoelectric emission is not possible, regardless of the intensity of light. **Q11:** The minimum negative (retarding) potential given to the anode with respect to the cathode for which the photoelectric current becomes zero. **Q12:** (a) Both A and R are true and R is the correct explanation of A. **Q13:** (a) Both A and R are true and R is the correct explanation of A. **Q14:** (a) Both A and R are true and R is the correct explanation of A. **Q15:** Photoelectric effect (or Compton effect). **Q16:** Zero. **Q17:** No, stopping potential is completely independent of the intensity of incident light. **Q18:** The electron. ($\lambda = \frac{h}{\sqrt{2mK}}$. Since K is same, smaller mass (electron) gives a larger wavelength). **Q19:** $\lambda = \frac{h}{\sqrt{2meV}} \approx \frac{1.227}{\sqrt{V}} \text{ nm}$. **Q20:** It remains entirely unchanged (saturation current depends only on intensity, not frequency). **Q21:** (i) Threshold frequency remains completely unchanged (it is an intrinsic property of the metal). (ii) Stopping potential increases linearly with the incident frequency (since K_{max} increases). **Q22:** Increasing intensity merely increases the number of photons, but not their individual energy. Only if a single photon's energy ($h\nu$) exceeds the work function will it eject an electron. Thus, a minimum frequency (threshold) is mandatory. **Q23:** (A) Maximum kinetic energy increases (since $K_{max} = h\nu - \Phi$). (B) Photocurrent remains unchanged (it depends only on the number of incident photons/intensity). **Q24:** $\frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{m_p q_p V}{m_\alpha q_\alpha V}} = \sqrt{\frac{m_p \times e}{4m_p \times 2e}} = \frac{1}{\sqrt{8}} = 1 : 2\sqrt{2}$. **Q25:** $\nu_{0A} > \nu_{0B} > \nu_{0C}$. For A, incident $\nu < \nu_{0A}$. For B, incident $\nu = \nu_{0B}$. For C, incident $\nu > \nu_{0C}$.

Since incident ν is the same, A has the highest threshold frequency and C has the lowest.

Q26: $N = \frac{P}{E} = \frac{2.0 \times 10^{-3}}{h\nu} = \frac{2.0 \times 10^{-3}}{6.63 \times 10^{-34} \times 6.0 \times 10^{14}} = 5.0 \times 10^{15} \text{ photons/s.}$ **Q27:**

$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.150 \times 30} = \frac{6.63 \times 10^{-34}}{4.5} = 1.47 \times 10^{-34} \text{ m.}$ **Q28:** $V_0 = \frac{hc}{e} \left(\frac{1}{\lambda} \right) - \frac{\Phi}{e}$. The slope of the line is $\frac{hc}{e}$. Knowing c and e , Planck's constant $h = \frac{\text{Slope} \times e}{c}$. **Q29:**

$\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA} = \frac{12.27}{\sqrt{50000}} \approx 0.055 \text{ \AA} \text{ (or } 0.0055 \text{ nm).}$ **Q30:** Graph is a straight line passing through the origin. $\lambda = \frac{h}{\sqrt{2mqV}} \Rightarrow \lambda\sqrt{V} = \frac{h}{\sqrt{2mq}}$. The slope is directly related to

$1/\sqrt{q/m}$. **Q31:** Since $\nu_1 > \nu_2$, the energy of photons in the first radiation ($h\nu_1$) is greater.

From $K_{max} = h\nu - \Phi$, the radiation with frequency ν_1 will eject electrons with a higher maximum kinetic energy. **Q32:** $\lambda = \frac{h}{mv} \Rightarrow v = \frac{h}{m\lambda}$. $\frac{v_p}{v_\alpha} = \frac{m_\alpha}{m_p} = \frac{4}{1}$. **Q33:** $K = h\nu_1 - h\nu_0$

and $2K = h\nu_2 - h\nu_0$. Multiplying the first by 2:

$2h\nu_1 - 2h\nu_0 = h\nu_2 - h\nu_0 \Rightarrow h\nu_0 = 2h\nu_1 - h\nu_2 \Rightarrow \nu_0 = 2\nu_1 - \nu_2$. **Q34:** Energy of photon $E_{ph} = \frac{hc}{\lambda}$. KE of electron $K_e = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$. Ratio $\frac{E_{ph}}{K_e} = \left(\frac{hc}{\lambda} \right) / \left(\frac{h^2}{2m\lambda^2} \right) = \frac{2mc\lambda}{h}$.

Thus $E_{ph} = \frac{2\lambda mc}{h} K_e$. **Q35:** (A) Photocurrent remains absolutely unchanged (graph is a horizontal straight line). (B) KE strictly increases linearly but more than doubles (

$K_2 = 2h\nu - \Phi = 2(K_1 + \Phi) - \Phi = 2K_1 + \Phi$). Graph is a straight line with a positive y-intercept. **Q36:** (A) $E = h\nu = \frac{6.63 \times 10^{-34} \times 6.4 \times 10^{14}}{1.6 \times 10^{-19}} \approx 2.65 \text{ eV.}$ (B)

$K_{max} = 2.65 - 2.31 = 0.34 \text{ eV.}$ (C) Stopping potential $V_0 = 0.34 \text{ V.}$ **Q37:** $K_1 = \frac{hc}{\lambda_1} - \Phi$

and $2K_1 = \frac{hc}{\lambda_2} - \Phi$. $2\left(\frac{hc}{\lambda_1} - \Phi\right) = \frac{hc}{\lambda_2} - \Phi \Rightarrow \Phi = hc\left(\frac{2}{\lambda_1} - \frac{1}{\lambda_2}\right)$. Threshold wavelength $\lambda_0 = \frac{hc}{\Phi} = \frac{\lambda_1\lambda_2}{2\lambda_2 - \lambda_1}$. **Q38:** Wave theory fails because it predicts that highly intense low-

frequency light should eventually eject electrons (violating the threshold frequency rule), and that energy absorption should take time (violating the instantaneous emission rule).

Einstein's $K_{max} = h\nu - \Phi$ explains both perfectly. **Q39:** $E_{inc} = \frac{1242 \text{ eV nm}}{412.5 \text{ nm}} \approx 3.01 \text{ eV.}$

Emission occurs only if $E_{inc} > W$. Thus, Na (1.92 eV) and K (2.15 eV) will show emission. Ca and Mo will not.

Q40: $K_{max} = h\nu - \Phi \Rightarrow \frac{1}{2}mv_{max}^2 = h\nu - \Phi \Rightarrow v_{max}^2 = \left(\frac{2h}{m}\right)\nu - \left(\frac{2\Phi}{m}\right)$.

Slope $l = \frac{2h}{m} \Rightarrow h = \frac{ml}{2}$. Y-intercept $n = -\frac{2\Phi}{m} \Rightarrow \Phi = -\frac{mn}{2}$ (where n is negative). **Q41:**

(A) No, they range from zero to K_{max} due to energy loss in internal collisions. (B) No, it

depends only on frequency. (C) It depends strictly on the intensity of the incident light

(number of photons per unit area per second). **Q42:**

$\nu_0 = 8 \times 10^{14} \text{ Hz} \Rightarrow \Phi = h\nu_0 = \frac{6.63 \times 10^{-34} \times 8 \times 10^{14}}{1.6 \times 10^{-19}} \approx 3.31 \text{ eV.}$ $K_{max} = 3.3 \text{ eV.}$

$E = K_{max} + \Phi = 3.3 + 3.31 = 6.61 \text{ eV.}$ $\nu = \frac{E}{h} = \frac{6.61 \times 1.6 \times 10^{-19}}{6.63 \times 10^{-34}} \approx 1.59 \times 10^{15} \text{ Hz.}$

Q43: (A) Beam A has a higher saturation current, thus it has a higher intensity. (B) The beam

with the largest negative stopping potential (most negative x-intercept) has the highest

frequency, because $eV_0 = h\nu - \Phi$. (C) Maximum momentum $p = \sqrt{2mK_{max}}$; the beam

with the highest stopping potential has the max KE, hence max momentum. (D) Doubling

intensity has no effect on stopping potential. **Q44:** (A) Equating photon energy to work

function plus max KE: $h\nu = \Phi + K_{max}$. (B) Photons travel at c , have zero rest mass, and

energy $E = h\nu$. (C) For P, $10^{15} \text{ Hz} < \nu_0$. For Q, $10^{15} \text{ Hz} = \nu_0$, so $K_{max} = 0$. Work function

$\Phi_Q = h(10^{15}) = \frac{6.63 \times 10^{-34} \times 10^{15}}{1.6 \times 10^{-19}} = 4.14 \text{ eV.}$ **Q45:** (A) Material particles in motion display

wave-like properties; $\lambda = \frac{h}{p}$. (B) Slope is proportional to $1/\sqrt{m}$. The steeper line represents

the smaller mass (m_2). (C) $\lambda = \frac{h}{\sqrt{2mK}}$. Since $m_\alpha > m_p$ and K is the same, the alpha particle has the smaller (shorter) wavelength. **Q46:** (A) Threshold frequency: min freq for emission. Stopping potential: min negative potential to stop photocurrent. (B) (i) $E = \frac{1242}{200} = 6.21 \text{ eV}$. $K_{max} = 6.21 - 4.2 = 2.01 \text{ eV}$. (ii) Zero change (intensity does not affect KE). (iii) $6.21 \text{ eV} < 6.5 \text{ eV}$, so emission does not occur (zero energy). **Q47:** (A) Intensity is the number of photons crossing a unit area per second multiplied by the energy of each photon ($I = n \cdot h\nu$). (B) No, rest mass is strictly zero. Momentum $p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$. (C) $E = h\nu = 6.63 \times 10^{-34} \times 6.0 \times 10^{14} \approx 3.98 \times 10^{-19} \text{ J}$. Number of photons $n = \frac{P}{E} = \frac{2.0 \times 10^{-3}}{3.98 \times 10^{-19}} \approx 5.0 \times 10^{15} \text{ s}^{-1}$. **Q48:** (A) Draw two curves starting at the same $-V_0$ but plateauing at different positive saturation current levels. (B) Draw two parallel lines. The negative y-intercept gives $-\Phi/e$. (C) $K_1 = h(2f) - hf = hf \Rightarrow v_1^2 \propto hf$. $K_2 = h(5f) - hf = 4hf \Rightarrow v_2^2 \propto 4hf$. Ratio $v_1^2/v_2^2 = 1/4 \Rightarrow v_1 : v_2 = 1 : 2$. **Q49:** (A) $p = \sqrt{2mK}$, so $\lambda = \frac{h}{\sqrt{2mK}}$. (B) Since $\lambda \propto 1/\sqrt{m}$ for constant K , the electron (having the smallest mass) has the largest wavelength. (C) $\lambda = 590 \text{ nm}$. $K = \frac{h^2}{2m\lambda^2} = \frac{(6.63 \times 10^{-34})^2}{2 \times 9.1 \times 10^{-31} \times (590 \times 10^{-9})^2} \approx 6.9 \times 10^{-24} \text{ J} = 4.3 \times 10^{-5} \text{ eV}$. **Q50:** (A) Stopping potential is independent of intensity; it remains constant. (B) Incident energy $E = \frac{1242 \text{ eV nm}}{330 \text{ nm}} = 3.76 \text{ eV}$. Metals with $W > 3.76 \text{ eV}$ will not emit. Thus, Mo (4.17 eV) and Ni (5.15 eV) will not cause emission. (C) Bringing it nearer only increases intensity (number of photons), not the energy of individual photons. Mo and Ni will *still* not emit electrons.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).